

Week 4 Preparation:**Student:** Jaskaran Singh

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Role: Data & Labelling Lead**Project:** ShieldSense - Accessible Scam-Message Screening Application**Week 4 Checkpoint:** Checkpoint 1 - continue / narrow / redirect**1. One Artefact Produced This Week:**

I have produced three linked artefacts this week, all of which sit under my role as Data & Labelling Lead.

Artefact 1 - Refined risk-category taxonomy (version 2): This is the second iteration based on feedback from the Week 3 meeting. It includes:

- Three final risk categories: Low, Medium, High.
- A clear definition for each category.
- Two examples per category, plus edge cases.
- A decision tree for ambiguous messages (e.g., messages that sit between Low and Medium).
- Notes on hard negatives such as legitimate service notifications from banks, Australia Post, and telcos.

Artefact 2 - Finalised labelling rules: This is the companion document all four labellers will use. It includes:

- The taxonomy definitions in operational terms.
- How to label edge cases and partial matches.
- The adjudication process when two labellers disagree.
- The process for the 30 double-labelled messages used to measure inter-rater agreement.
- The rationale for the metric I have chosen (see decision below).

Artefact 3 - Label consistency check log. This is the record of the consistency check I ran on the first batch of labelled messages. It includes:

- The first 50 messages labelled by me and cross-checked against the labelling rules.
- Notes on where I had to pause and re-read the rules because a message was ambiguous.
- The cases where I flagged a label for a second opinion before finalising it.
- A short summary of patterns I noticed - for example, messages that looked risky but were legitimate service notifications, and messages that looked harmless but contained subtle urgency cues.

This consistency log is important because it becomes the evidence that our labels are not just applied but checked, and it feeds directly into the inter-rater agreement report I will complete next week.

All three documents are now committed to our team GitHub repository under my own account, since we are all set up on GitHub as of this week.

Evidence: Committed taxonomy v2, labelling rules, and consistency check log in the team GitHub repository under my own account; printed copies brought to the meeting for review.

2. One Decision I Need from You:

I need your ruling on the held-out test set split ratio.

I am preparing to split off the held-out test set from the 150–250 messages. My plan is:

- **Option A:** 80/20 split (approximately 200 training, 50 test)
- **Option B:** 70/30 split (approximately 175 training, 75 test)
- **Option C:** Stratified split by risk category, so each risk band is proportionally represented in both training and test sets.

I lean towards Option C (a stratified split) because it guarantees the test set reflects the real distribution of Low, Medium and High risk messages, which is better for honest evaluation. But I want your ruling because it affects how the Model Lead evaluates the classifier against the rules baseline, and I want to make sure this aligns with your expectations for the comparison table.

Second, I also want to confirm the inter-rater agreement metric. At Week 3 I proposed using Cohen's kappa as the primary metric with percentage agreement as a secondary check. I want to confirm this is still your preferred approach before I finalise the agreement section of the data card.

Why I cannot resolve these alone: The split ratio affects the quality of the comparison table, which is a client deliverable. The agreement metric affects how dataset quality is evidenced for the Model Lead's evaluation and for the client handover. Both need your ruling at the meeting.

2. Progress Against My Role This Week:

What I have completed:

- Refined taxonomy to version 2 based on Week 3 feedback.
- Finalised the labelling rules document.
- Began the dataset labelling process.
- Ran a consistency check on the first 50 messages and logged the results.
- Committed all three artefacts to GitHub under my own account.
- Shared the agreed taxonomy and dataset structure with the Model & Evaluation Lead.

What is in progress:

- Completing the 150–250 verified messages.
- Splitting off and locking the held-out test set (pending your ruling on the split ratio).
- Drafting the data card.
- Running the 30 double-labelled messages for inter-rater agreement.

What is at risk:

- If the split ratio decision is delayed, I cannot lock the test set by Friday, which will push the Model Lead's training timeline into Week 5.
- I am also aware this is Checkpoint 1 — continue / narrow / redirect. I want to hear your ruling on whether the scope is still achievable and whether I should narrow any part of it.

5. Blocker I Am Raising:

The main blocker is the held-out test set split ratio decision. I need your ruling on 80/20, 70/30, or stratified split before I lock the test set. This affects the Model Lead's training timeline and the quality of the comparison table.

The GitHub repository blocker from Week 3 is now resolved - we are all set up and I have committed my work under my own account this week.

Everything else is on track for the Week 4 milestone.

6. Checkpoint 1 My Assessment:

This is the Week 4 checkpoint, so I want to give you an read on where I am:

- The taxonomy and labelling rules are strong. They are ready for the dataset sprint and have been shared with the Model Lead.
- The dataset is progressing but not complete. I expect to finish labelling by Sunday.
- The consistency check is done on the first batch and has already surfaced useful patterns - for example, legitimate service notifications that initially looked risky.
- The test set is not yet locked because I am waiting on the split ratio decision.
- My recommendation for the checkpoint: Continue, with one narrowing - I would like to confirm the split ratio decision this week so I can lock the test set on time. I do not think redirect is needed, but I want your ruling.

7. Next Week (Week 5) - What I Will Bring:

By next Thursday I will bring:

- The completed 150–250 message dataset with verified labels.
- The held-out test set locked and documented.
- The draft data card.
- The inter-rater agreement report from the 30 double-labelled messages.
- The final consistency check summary across the full dataset.

- My updated logbook.
- Evidence of GitHub commits under my own account.

Week 5 is also the clickable demo week, so I will support the Model Lead and Interface Lead with any dataset examples they need for the Green/Amber/Red explanation flow.